Blue Team Tools Resource Guide

Windows and Linux Tools for System Administration, Defense, and Monitoring

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Online Resources and Tools

Portmaster (Safing)

What it is: Free, open-source application firewall and network monitor for Windows. It gives per-application network control, DNS-level ad/tracker blocking, and real-time connection visibility. Think of it as a modern, GUI-based replacement for Windows Firewall with application-level granularity.

Website: https://safing.io/portmaster

Download: https://safing.io/download

Why it matters for CCDC: Portmaster lets you see exactly which processes are making network connections and to where, block unauthorized outbound traffic per-application, and monitor for suspicious network activity in real time. It is far more usable than raw `netsh` firewall rules during a fast-paced competition.

Installation

1. Download the installer from https://safing.io/download (approximately 50 MB).
2. Run the installer as Administrator. Default install path is `C:\Program Files\Safing\Portmaster\`.
3. Portmaster runs as a system service and starts automatically.
4. Access the UI by clicking the system tray icon or navigating to `http://127.0.0.1:817/` in a browser.

Install methods:

- **GUI installer:** Download from https://safing.io/download (~50 MB)
- **Winget:** `winget install -i portmaster`
- **Offline installer:** Build one using https://github.com/innovatodev/PortmasterOffline (~300 MB, bundles all dependencies). Critical for competition environments without internet.

Install path: `C:\ProgramData\Safing\Portmaster\`

Uninstall: `C:\ProgramData\Safing\Portmaster\portmaster-uninstaller.exe`

No account creation is required. Portmaster is fully functional without an account. Safing offers a paid "SPN" (Safing Privacy Network) VPN service, but the firewall, network monitor, and DNS features are all free. Uses Windows Filtering Platform (WFP) kernel driver for packet-level interception.

Key Features for Blue Teams

Feature	How It Helps
Per-app network rules	Block specific processes from reaching the internet (e.g., block `powershell.exe` outbound)
Connection history	See every connection each application has made, with timestamps and destination IPs
DNS monitoring	See all DNS queries with the requesting process. Detect C2 beaconing by domain
Bulk allow/deny	Set global defaults (deny all, allow list) or per-app policies
Bandwidth monitor	Identify processes consuming unusual bandwidth (data exfiltration)
System service	Runs as `PortmasterCore` service, survives user logoff

Quick Configuration for Competition

1. **Set global default to "Prompt"** – forces you to approve new outbound connections.
2. **Allow scored services** – HTTP/HTTPS, DNS, SSH, RDP, whatever your box serves.
3. **Block PowerShell and cmd.exe outbound** – prevents reverse shells from calling home.
4. **Monitor the connection log** – look for unexpected destinations or high connection counts.

Useful Portmaster CLI

```
# Check service status
Get-Service PortmasterCore

# Restart Portmaster
Restart-Service PortmasterCore

# Logs location
Get-ChildItem "$env:ProgramData\Safing\Portmaster\logs\"
```

---

runZero (Formerly Rumble)

What it is: Network asset discovery and inventory tool. Scans your network and builds a detailed inventory of every device, OS, open port, and service – without agents. The free Community Edition supports up to 256 assets.

Website: https://www.runzero.com

Download: https://www.runzero.com/download

Account: Free account required at https://console.runzero.com/signup

Why It Matters for CCDC

At the start of competition, you inherit an unknown network. runZero can scan the entire subnet and produce a complete inventory in minutes – OS versions, open ports, services, hostnames, MAC addresses. This feeds directly into your inventory sheets and network diagram.

Quick Start

1. Create a free account at https://console.runzero.com/signup
2. Download the Explorer (scanner agent) for your platform
3. Deploy the Explorer on one host with network access to all subnets
4. Run a scan of your competition subnet(s)
5. Export the results as CSV for your inventory sheets

Key Scan Output

- Hostname, IP, MAC address
- Operating system and version
- Open TCP/UDP ports and service banners
- SSL/TLS certificate details
- Hardware vendor (from MAC OUI)
- Detected vulnerabilities

---

simplewall (Lightweight Alternative to Portmaster)

What it is: Free, open-source, portable firewall under 2 MB. Uses WFP directly (independent of Windows Firewall). Includes built-in Windows telemetry blocklist. No installation required – just run the executable.

Website: https://simplewall.net

GitHub: https://github.com/henrypp/simplewall

When to use instead of Portmaster: When you need a smaller, faster-to-deploy firewall on boxes where Portmaster's 300 MB offline installer is impractical. simplewall is a single portable EXE.

Online Resources Reference

Threat Intelligence and IOC Lookup

Resource	URL	Purpose
VirusTotal	https://www.virustotal.com	Scan files, hashes, URLs, domains against 70+ AV engines
AbuseIPDB	https://www.abuseipdb.com	Check/report abusive IP addresses (1000 free queries/day)
Cisco Talos	https://www.talosintelligence.com/reputation_center	IP, domain, and URL reputation lookup
GreyNoise	https://www.greynoise.io	Identify mass-scanning IPs vs. targeted attacks
Shodan	https://www.shodan.io	Search engine for internet-exposed services and ports
Censys	https://search.censys.io	Host/service search with certificate and vuln data
URLhaus	https://urlhaus.abuse.ch	Database of malicious URLs used for malware distribution
URLScan.io	https://urlscan.io	Analyze what a website loads, where it redirects
Malware Bazaar	https://bazaar.abuse.ch	Malware sample sharing and hash lookup
ThreatFox	https://threatfox.abuse.ch	IOC sharing platform (hashes, IPs, domains)
AlienVault OTX	https://otx.alienvault.com	Open threat intelligence with 19M+ indicators
Pulsedive	https://pulsedive.com	Free threat intel enriching IPs, domains, URLs

Vulnerability Databases

Resource	URL	Purpose
NIST NVD	https://nvd.nist.gov	U.S. government CVE database with severity scores
CVE.org	https://www.cve.org	Official CVE identifier lookup
CVE Details	https://www.cvedetails.com	CVE database with advisories, exploits, vendor browsing
CISA KEV Catalog	https://www.cisa.gov/known-exploited-vulnerabilities-catalog	Vulns actively exploited in the wild – patch these first
Exploit-DB	https://www.exploit-db.com	Public exploit archive and Google Hacking Database
Rapid7 Vuln DB	https://www.rapid7.com/db	180K+ vulns with Metasploit module references

Security Configuration Guides

Resource	URL	Purpose
CIS Benchmarks	https://www.cisecurity.org/cis-benchmarks	Free PDF hardening guides for OSes, servers, cloud
DISA STIGs	https://public.cyber.mil/stigs	DoD hardening guides, freely downloadable
MS Security Baselines	https://learn.microsoft.com/en-us/windows/security/operating-system-security/device-management/windows-security-configuration-framework/windows-security-baselines	Recommended GPO settings for Windows
MS Security Compliance Toolkit	https://www.microsoft.com/en-us/download/details.aspx?id=55319	Download, compare, apply MS security baselines
NIST Checklists	https://nvd.nist.gov/ncp	800+ security config checklists for 200+ systems
MITRE ATT&CK	https://attack.mitre.org	Adversary tactics and techniques knowledge base

Log Analysis and Decoding

Resource	URL	Purpose
CyberChef	https://gchq.github.io/CyberChef	Decode Base64, analyze payloads, convert formats
Regex101	https://regex101.com	Online regex tester with real-time explanation
Ultimate Win Security	https://www.ultimatewindowssecurity.com/securitylog/encyclopedia/	Windows Security Event ID encyclopedia
MyEventLog	https://www.myeventlog.com	Windows event log message database
MS Events to Monitor	https://learn.microsoft.com/en-us/windows-server/identity/ad-ds/plan/appendix-l–events-to-monitor	Critical AD security events list
awesome-event-ids	https://github.com/stuhli/awesome-event-ids	Curated important Windows Event IDs
Win Event Log Analyst PDF	https://forwarddefense.com/media/attachments/2021/05/15/windows-event-log-analyst-reference.pdf	Windows event log analysis reference

Network Tools

Resource	URL	Purpose
MXToolbox	https://mxtoolbox.com	DNS, MX, blacklist, SMTP diagnostics
NSLookup.io	https://www.nslookup.io	Web-based DNS client for all record types
DNS Checker	https://dnschecker.org	DNS propagation check plus 30+ network tools
SSL Labs	https://www.ssllabs.com/ssltest	Deep SSL/TLS config analysis, grades A-F
NetworkCalc	https://networkcalc.com	Subnet calculator, DNS, WHOIS, SSL tools
ICANN WHOIS	https://lookup.icann.org	Official domain registration lookup

Malware Analysis

Resource	URL	Purpose
ANY.RUN	https://any.run	Interactive malware sandbox with real-time observation
Hybrid Analysis	https://hybrid-analysis.com	Free automated malware analysis (CrowdStrike)
Joe Sandbox	https://www.joesandbox.com	Automated analysis on Windows, macOS, Linux
YARAify	https://yaraify.abuse.ch	Scan files against YARA rule repository online
Yara-Rules	https://github.com/Yara-Rules/rules	Community YARA signature repository

Documentation and Reference

Resource	URL	Purpose
SS64	https://ss64.com	A-Z command reference for CMD, PowerShell, Bash
ExplainShell	https://explainshell.com	Visual breakdown of any shell command
TLDR Pages	https://tldr.sh	Simplified, example-driven man pages
MS PowerShell Docs	https://learn.microsoft.com/en-us/powershell	Official PowerShell cmdlet reference
Sysinternals Live	https://live.sysinternals.com	Run Sysinternals tools directly from web

CCDC Community Resources

Resource	URL	Purpose
awesome-cybersecurity-blueteam	https://github.com/fabacab/awesome-cybersecurity-blueteam	Curated blue team tools and frameworks
A-poc/BlueTeam-Tools	https://github.com/A-poc/BlueTeam-Tools	65+ IR and blue team tools
awesome-security-hardening	https://github.com/decalage2/awesome-security-hardening	Hardening guides, benchmarks, checklists
C0nd4/CCDC-Blueteam-Manual	https://github.com/C0nd4/CCDC-Blueteam-Manual	Blue team manual for CCDC
Missouri-State CCDC	https://github.com/Missouri-State-CCDC-Team/mwccdc	Splunk dashboards, Ansible, templates
CyberLions/CCDC	https://github.com/CyberLions/CCDC	Penn State CCDC scripts and tools
GitHub blue-team topic	https://github.com/topics/blue-team	Browse all blue-team tagged repos

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Microsoft Sysinternals Suite

Download the full suite: https://learn.microsoft.com/en-us/sysinternals/downloads/sysinternals-suite

Live web access (no download): `\\live.sysinternals.com\tools\` or https://live.sysinternals.com

Deploy to multiple machines: Copy the extracted suite folder to a network share or SYSVOL, then run tools directly from `\\server\share\SysinternalsSuite\`.

Install methods:

- **Direct download (ZIP):** https://download.sysinternals.com/files/SysinternalsSuite.zip (~45 MB)
- **Chocolatey:** `choco install sysinternals`
- **Winget:** `winget install Microsoft.Sysinternals.Suite`
- **UNC live path:** `net use S: \\live.sysinternals.com\DavWWWRoot` (requires WebDAV client service)

All tools are portable (no installation required). Extract and run. Always use `-accepteula` on first run to suppress the EULA dialog in scripts.

Process and System Monitoring

Process Explorer (`procexp64.exe`)

What: Advanced Task Manager replacement. Shows process tree, DLLs loaded by each process, handles, TCP/IP connections, GPU usage, and VirusTotal integration.

Blue team use: Identify suspicious processes, find hidden malware injected into legitimate processes, see which DLLs a process has loaded, check process signatures.

```
procexp64.exe /t          # Start minimized to tray
procexp64.exe /e          # Start with Everyone access
```

Key actions:

- **Options > VirusTotal.com > Check VirusTotal.com** – submits hashes of all running processes
- **View > Show Lower Pane** – shows DLLs/handles for selected process
- **Find > Find Handle or DLL** – search for which process has a file/registry key open
- Right-click process > **Properties > TCP/IP** – see network connections for that process
- **Ctrl+D** = DLL view, **Ctrl+H** = Handle view
- **Color codes:** Purple = packed/encrypted image (suspicious), Pink = services, Blue = your own processes, Dark gray = suspended

Process Monitor (`procmon64.exe`)

What: Real-time file system, registry, and process/thread activity monitor. Captures every operation with full call stacks.

Blue team use: Watch what a suspicious process is doing – what files it reads/writes, what registry keys it touches, what network connections it makes. Essential for malware analysis.

```
procmon64.exe /AcceptEula /BackingFile C:\logs\procmon.pml
procmon64.exe /Quiet /Minimized /BackingFile C:\logs\procmon.pml
```

Key filters:

- **Process Name is** `powershell.exe` – watch PowerShell activity
- **Operation is** `WriteFile` – see all file writes
- **Path contains** `\Run\` – watch autorun registry keys
- **Result is** `ACCESS DENIED` – find permission issues

Autoruns (`autoruns64.exe` / `autorunsc64.exe`)

What: Shows everything configured to run at startup or login. Covers registry Run keys, services, scheduled tasks, drivers, Winsock providers, DLLs, boot execute entries, image hijacks, and more. The most comprehensive autostart viewer available.

Blue team use: Find persistence mechanisms planted by red team. Compare against a known-good baseline.

```
:: GUI version
autoruns64.exe

:: CLI version -- export to CSV for diffing
autorunsc64.exe -a * -c -h -s -v -vt > C:\logs\autoruns.csv

:: Compare two snapshots
:: (Use File > Compare in GUI to diff against saved .arn file)
```

Flags for `autorunsc64.exe`:

Flag	Purpose
`-a *`	Show all categories
`-c`	CSV output
`-h`	Show file hashes
`-s`	Verify digital signatures
`-v`	Verify with VirusTotal
`-vt`	Accept VirusTotal TOS
`-m`	Hide Microsoft-signed entries

Startup categories checked: Logon, Explorer, Internet Explorer, Services, Drivers, Scheduled Tasks, AppInit DLLs, Boot Execute, Image Hijacks, KnownDLLs, Winsock Providers, Print Monitors, LSA Providers, Network Providers, WMI, Office, Sidebar Gadgets, Codecs.

Sysmon (`sysmon64.exe`)

What: System Monitor service that logs detailed process creation, network connections, file creation time changes, driver/image loads, raw disk access, process memory access, and more to the Windows Event Log.

Blue team use: Provides the detailed telemetry needed to detect attacker activity. Once installed with a config, it continuously logs to `Microsoft-Windows-Sysmon/Operational`.

```
:: Install with config
sysmon64.exe -accepteula -i sysmonconfig.xml

:: Update config
sysmon64.exe -c sysmonconfig.xml

:: Uninstall
sysmon64.exe -u
```

Recommended configs:

- SwiftOnSecurity: https://github.com/SwiftOnSecurity/sysmon-config
- Olaf Hartong: https://github.com/olafhartong/sysmon-modular

Key Event IDs:

Event ID	What It Logs
1	Process creation (with command line, hashes, parent process)
3	Network connection (source/dest IP, port, process)
7	Image loaded (DLL loading)
8	CreateRemoteThread (process injection)

Event ID	What It Logs
10	ProcessAccess (credential dumping, e.g., LSASS access)
11	FileCreate
13	RegistryEvent (value set)
2	File creation time changed (timestomping detection)
6	Driver loaded (rootkit driver detection)
22	DNSEvent (DNS query with process)
23	File delete archived
25	Process tampering

Logs location: Event Viewer > Applications and Services Logs > Microsoft > Windows > Sysmon > Operational

ProcDump (`procdump64.exe`)

What: Command-line crash dump utility. Captures memory dumps of processes based on triggers (CPU threshold, hang, exception, or manual).

Blue team use: Dump a suspicious process's memory for offline analysis before killing it. Preserve evidence of malware.

```
:: Full memory dump of suspicious process
procdump64.exe -accepteula -ma <PID> C:\evidence\

:: Dump when CPU exceeds 90%
procdump64.exe -accepteula -ma -c 90 <PID> C:\evidence\

:: Dump on process termination (catch what it does last)
procdump64.exe -accepteula -ma -t <PID> C:\evidence\
```

BgInfo (`bginfo64.exe`)

What: Sets desktop wallpaper to display system information: hostname, IP address, OS version, domain, uptime, etc.

Blue team use: When RDP'ing between 8+ machines during competition, BgInfo lets you instantly identify which host you are on. Deploy to all machines at competition start.

```
:: Apply config silently (create .bgi file in GUI first, then deploy)
bginfo64.exe config.bgi /timer:0 /nolicprompt /silent
```

Network Tools

TCPView (`tcpview64.exe`)

What: GUI showing all TCP and UDP endpoints with owning process, state, and bytes transferred. Real-time updates.

Blue team use: Instantly see which processes have active network connections and to where. Spot reverse shells, C2 beacons, and unauthorized listeners.

```
tcpview64.exe
```

Key features:

- Red = new connection, Yellow = state change, Green = closing
- Right-click > End Process to kill suspicious connections
- Right-click > Close Connection to reset a specific TCP connection
- File > Save to export current state

CLI version (`tcpvcon64.exe`) for scripting:

```
tcpvcon64.exe -accepteula -a -c -n > C:\baseline-connections.csv
```

PsPing (`psping64.exe`)

What: TCP ping, latency, and bandwidth measurement tool. Tests TCP connectivity to specific ports (unlike ICMP ping which only tests basic reachability).

Blue team use: Test if a service port is actually reachable through firewalls.

```
psping64.exe -t server:3389          # TCP ping to RDP port
psping64.exe -t server:443 -i 5     # Ping every 5 seconds
```

Security and Access Tools

AccessChk (`accesschk64.exe`)

What: Shows effective permissions for users/groups on files, directories, registry keys, services, and kernel objects.

Blue team use: Find weak permissions on services (unquoted paths, writable service binaries), world-writable directories, and misconfigured registry keys.

```
:: Check permissions on all services
accesschk64.exe -accepteula -ucqv *

:: Find services writable by Authenticated Users
accesschk64.exe -accepteula -uwcqv "Authenticated Users" *

:: Check write access to a directory
accesschk64.exe -accepteula -uwdq "C:\Program Files"

:: Check registry key permissions
accesschk64.exe -accepteula -k HKLM\Software
```

AccessEnum (`AccessEnum.exe`)

What: GUI tool that shows permissions on directories, files, and registry keys, highlighting where permissions diverge from parent.

Blue team use: Quickly audit file system permissions to find where non-admin users have write access to sensitive directories.

Sigcheck (`sigcheck64.exe`)

What: Verifies digital signatures on files, shows version info, and can submit hashes to VirusTotal.

Blue team use: Verify that system binaries have valid Microsoft signatures. Find unsigned or tampered executables.

```
:: Check all unsigned EXEs in System32
sigcheck64.exe -accepteula -u -e C:\Windows\System32

:: Check all unsigned EXEs recursively
sigcheck64.exe -accepteula -u -s -e C:\Windows

:: Submit to VirusTotal
sigcheck64.exe -accepteula -v -vt C:\suspicious\file.exe
```

LogonSessions (`logonsessions64.exe`)

What: Lists all active logon sessions with their type, authentication package, logon time, and username.

Blue team use: Find unexpected logon sessions (red team lateral movement, pass-the-hash).

```
logonsessions64.exe -accepteula -p    # Include processes per session
```

ShareEnum (`ShareEnum.exe`)

What: Scans the network for shared folders and shows their permissions.

Blue team use: Find open shares with weak ACLs that could be used for lateral movement or data exfiltration.

Handle (`handle64.exe`)

What: Shows which processes have open handles to files, registry keys, or other objects.

Blue team use: Find which process is locking a file (cannot delete red team tool because something holds it open). Find processes with handles to sensitive files like SAM or SYSTEM hive.

```
handle64.exe -accepteula -p suspicious.exe
handle64.exe -accepteula SAM               # Who has SAM open?
```

ListDLLs (`listdlls64.exe`)

What: Lists all DLLs loaded by a process, including load addresses and version info.

Blue team use: Detect DLL injection or sideloading. Look for DLLs loaded from unusual paths (Temp, AppData, Downloads).

```
:: Show all unsigned DLLs across all processes
listdlls64.exe -accepteula -u

:: Search for processes that loaded a specific DLL
listdlls64.exe -accepteula -d evil.dll
```

Strings (`strings64.exe`)

What: Extracts printable ASCII and Unicode strings from binary files.

Blue team use: Quick triage of suspicious binaries. Look for URLs, IP addresses, command strings, registry paths.

```
strings64.exe -accepteula suspicious.exe | findstr /i "http:// https:// cmd powershell"
```

File and Disk Tools

Streams (`streams64.exe`)

What: Shows NTFS alternate data streams (ADS) on files. ADS can hide data or executables within normal files.

Blue team use: Detect hidden payloads in alternate data streams.

```
:: Scan a directory recursively
streams64.exe -accepteula -s C:\Users\

:: Delete all ADS in a directory
streams64.exe -accepteula -s -d C:\Users\Public\
```

SDelete (`sdelete64.exe`)

What: Secure file deletion (overwrite before delete) and free-space wiping.

Blue team use: Securely remove sensitive files (PII found during competition) or wipe temp files attackers may have left.

```
sdelete64.exe -accepteula -p 1 C:\temp\malware.exe
sdelete64.exe -accepteula -z C:           # Zero free space
```

Du (`du64.exe`)

What: Shows disk usage by directory, sorted by size.

Blue team use: Find unexpectedly large directories (data staging areas, exfiltration caches).

```
du64.exe -accepteula -l 2 C:\Users       # Depth 2
```

Disk2vhd (`disk2vhd64.exe`)

What: Creates VHD/VHDX virtual disk images from live physical disks using Volume Shadow Copy.

Blue team use: Forensic imaging of a compromised machine without taking it offline. Image first, analyze later.

```
disk2vhd64.exe -accepteula * C:\evidence\image.vhdx
```

System Information and Remote Tools

PsExec (`psexec64.exe`)

What: Execute processes remotely on other Windows machines via SMB (admin shares).

Blue team use: Push commands, scripts, or tools to multiple machines without RDP. Deploy Sysmon, run Autoruns, push firewall rules.

```
:: Run command on remote machine
psexec64.exe \\server -u DOMAIN\admin -p password cmd /c ipconfig

:: Run on multiple machines
psexec64.exe \\server1,server2 -u admin -p pass -c autoruns64.exe

:: Copy and execute local tool on remote machine
psexec64.exe \\server -c -f sysmon64.exe -accepteula -i config.xml

:: Run interactively
psexec64.exe \\server -i -d cmd.exe

:: Run as SYSTEM locally (useful for testing)
psexec64.exe -s cmd.exe
```

PsInfo (`psinfo64.exe`)

What: Shows system information: OS version, uptime, install date, kernel build, processor, physical memory, hotfixes.

```
psinfo64.exe \\server -u admin -p pass
```

PsList (`pslist64.exe`)

What: Remote process lister (like `tasklist` but works over the network).

```
pslist64.exe \\server -u admin -p pass
pslist64.exe \\server -t        # Show process tree
```

PsService (`psservice64.exe`)

What: Remote service viewer/controller.

```
psservice64.exe \\server query          # List all services
psservice64.exe \\server stop MalSvc    # Stop a service
psservice64.exe \\server config MalSvc  # Show service config
```

PsLoggedOn (`psloggedon64.exe`)

What: Shows who is logged on locally and via resource shares.

```
psloggedon64.exe \\server
```

PsKill (`pskill64.exe`)

What: Kill a process by name or PID on a local or remote system.

```
pskill64.exe \\server powershell.exe
pskill64.exe \\server -t 1234          # Kill process tree
```

PsPasswd (`pspasswd64.exe`)

What: Change account passwords on local and remote systems. Works across multiple machines.

Blue team use: Mass password changes across all machines in the first 15 minutes.

```
:: Change local admin password on all domain machines
pspasswd64.exe \\* -u DOMAIN\admin -p oldpass Administrator N3wP@ssw0rd!

:: Change on specific machine
pspasswd64.exe \\DC01 Administrator N3wP@ssw0rd!
```

PsSuspend (`pssuspend64.exe`)

What: Suspend and resume processes without killing them.

Blue team use: Freeze a suspicious process to preserve its state for analysis (memory dump with ProcDump) before killing it.

```
pssuspend64.exe suspicious.exe          # Suspend
pssuspend64.exe -r suspicious.exe       # Resume
```

PsLogList (`psloglist64.exe`)

What: Dump event logs from local or remote systems to text.

Blue team use: Collect event logs from all machines to a central location for analysis.

```
:: Dump last 1000 Security events from remote machine
psloglist64.exe \\DC01 -n 1000 Security > DC01-security.txt

:: Dump from last 2 days
psloglist64.exe \\DC01 -d 2 Security > DC01-security-2d.txt
```

Tool	Purpose

Additional Useful Tools

Tool	Purpose
`Coreinfo.exe`	Show CPU/cache topology and security features (DEP, VT-x, NX)
`PendMoves64.exe`	Show pending file rename/delete operations (scheduled for reboot)
`Whois64.exe`	WHOIS lookup from command line
`vmmap64.exe`	Show virtual memory layout of a process
`RAMMap.exe`	Advanced physical memory usage analysis
`FindLinks64.exe`	Find hard links to a file (detect hardlink-based evasion)
`WinObj.exe`	Object Manager namespace browser (reveal hidden devices/drivers)

CCDC Priority Order

Deploy in this order at competition start:

1. **Sysmon** – Install with SwiftOnSecurity config on every box
2. **Autoruns** – Baseline all autostart locations, export CSV
3. **Sigcheck** – Scan System32 and user directories for unsigned executables
4. **PsPasswd** – Mass password change across all machines
5. **Process Explorer** – Visual triage of running processes, VirusTotal check
6. **TCPView** – Baseline network connections
7. **BgInfo** – Set on all machines for quick identification
8. **AccessChk** – Audit service permissions
9. **PsExec** – Primary remote administration tool
10. **LogonSessions** – Check for unauthorized sessions

Rapid Deployment Script

Deploy Sysinternals + Sysmon across multiple machines:

```
$tools = "\\DC01\NETLOGON\SysinternalsSuite"
$config = "\\DC01\NETLOGON\sysmonconfig.xml"
$targets = @("WEB01","DB01","MAIL01","DNS01")

foreach ($target in $targets) {
    # Copy tools
    Copy-Item $tools "\\$target\C$\Tools\Sysinternals" -Recurse -Force

    # Install Sysmon remotely
    Invoke-Command -ComputerName $target -ScriptBlock {
        & "C:\Tools\Sysinternals\sysmon64.exe" -accepteula -i "C:\Tools\sysmonconfig.xml"
    }

    # Run Autoruns baseline
    & "$tools\psexec64.exe" "\\$target" -accepteula `
        C:\Tools\Sysinternals\autorunsc64.exe -a * -c -h -s `
        > "C:\baselines\$target-autoruns.csv"
}
```

Windows Tools Beyond Sysinternals

Built-In Windows Security Commands

PowerShell Security Cmdlets

```
# ---- Process and Network ----
Get-Process | Sort-Object CPU -Descending | Select -First 20
Get-NetTCPConnection -State Established | Sort-Object RemotePort
Get-NetTCPConnection -State Listen
Get-NetUDPEndpoint

# ---- Scheduled Tasks (persistence) ----
Get-ScheduledTask | Where-Object {$_.State -eq "Ready"}
Get-ScheduledTask | Select TaskName,TaskPath,State | Format-Table -Auto
schtasks /query /fo CSV /v > C:\baseline\tasks.csv

# ---- Services ----
Get-Service | Where-Object {$_.Status -eq "Running"} | Sort-Object DisplayName
Get-WmiObject Win32_Service | Select Name,StartName,PathName,State | Export-Csv C:\baseline\services.csv

# ---- Installed Software ----
Get-ItemProperty HKLM:\Software\Microsoft\Windows\CurrentVersion\Uninstall\* |
    Select DisplayName,DisplayVersion,Publisher | Sort DisplayName

# ---- Local Users and Groups ----
Get-LocalUser | Select Name,Enabled,LastLogon
Get-LocalGroupMember -Group "Administrators"

# ---- AD Users and Groups (on DCs) ----
Get-ADUser -Filter * -Properties Enabled,LastLogonDate |
    Select Name,Enabled,LastLogonDate | Sort LastLogonDate
Get-ADGroupMember "Domain Admins" | Select Name
Get-ADGroupMember "Enterprise Admins" | Select Name

# ---- Firewall ----
Get-NetFirewallProfile | Select Name,Enabled
Get-NetFirewallRule -Enabled True -Direction Inbound |
    Select DisplayName,Action,Profile | Format-Table -Auto

# ---- Shares ----
Get-SmbShare | Select Name,Path,Description
Get-SmbShareAccess -Name "ShareName"

# ---- DNS Cache ----
Get-DnsClientCache | Select Entry,Data

# ---- Recently Modified Files ----
Get-ChildItem C:\Users -Recurse -File |
    Where-Object {$_.LastWriteTime -gt (Get-Date).AddHours(-24)} |
    Select FullName,LastWriteTime | Sort LastWriteTime -Descending
```

Windows Defender CLI (`MpCmdRun.exe`)

```
# Full path (run as admin)
$mp = "$env:ProgramFiles\Windows Defender\MpCmdRun.exe"
```

```
# Update signatures
& $mp -SignatureUpdate

# Quick scan
& $mp -Scan -ScanType 1

# Full scan
& $mp -Scan -ScanType 2

# Scan specific path
& $mp -Scan -ScanType 3 -File "C:\Users\Public"

# Check Defender status
Get-MpComputerStatus | Select AntivirusEnabled,RealTimeProtectionEnabled,
    AntivirusSignatureLastUpdated

# Add exclusion (if Defender is blocking a legitimate tool)
Add-MpPreference -ExclusionPath "C:\Tools"

# Remove exclusion
Remove-MpPreference -ExclusionPath "C:\Tools"

# List current exclusions (red team may have added some)
Get-MpPreference | Select ExclusionPath,ExclusionProcess,ExclusionExtension
```

auditpol.exe (Audit Policy)

```
# Show current audit policy
auditpol /get /category:*

# Enable full auditing (recommended for CCDC)
auditpol /set /category:"Account Logon" /success:enable /failure:enable
auditpol /set /category:"Account Management" /success:enable /failure:enable
auditpol /set /category:"Logon/Logoff" /success:enable /failure:enable
auditpol /set /category:"Object Access" /success:enable /failure:enable
auditpol /set /category:"Policy Change" /success:enable /failure:enable
auditpol /set /category:"Privilege Use" /success:enable /failure:enable
auditpol /set /category:"System" /success:enable /failure:enable
auditpol /set /category:"Detailed Tracking" /success:enable /failure:enable

# Export policy (baseline)
auditpol /backup /file:C:\baseline\auditpol.csv
```

wevtutil (Event Log Management)

```
# List all event logs
wevtutil el

# Export Security log to file
wevtutil epl Security C:\evidence\security.evtx

# Query recent failed logons (Event 4625)
wevtutil qe Security /q:"*[System[(EventID=4625)]]" /c:50 /f:text

# Clear a log (use with caution -- save first)
wevtutil cl Security

# Check log size
wevtutil gl Security
```

certutil (Certificate and Hash Operations)

```
# Hash a file (verify integrity)
certutil -hashfile C:\suspicious.exe SHA256

# Download a file (alternative to curl/wget)
certutil -urlcache -split -f http://example.com/file.exe C:\temp\file.exe

# List installed certificates
certutil -store My
certutil -store Root

# Decode Base64 file
certutil -decode encoded.txt decoded.bin
```

netsh (Firewall and Network)

```
# ---- Firewall ----
# Show all firewall rules
netsh advfirewall firewall show rule name=all

# Block an IP
netsh advfirewall firewall add rule name="Block Attacker" dir=in action=block remoteip=10.0.0.100

# Allow a port
netsh advfirewall firewall add rule name="Allow HTTP" dir=in action=allow protocol=tcp localport=80

# Export/import firewall rules (baseline and restore)
netsh advfirewall export C:\baseline\firewall.wfw
netsh advfirewall import C:\baseline\firewall.wfw

# Reset firewall to defaults
netsh advfirewall reset

# ---- Network Diagnostics ----
# Show all interfaces
netsh interface ipv4 show addresses

# Show ARP table
netsh interface ipv4 show neighbors

# Show routing table
netsh interface ipv4 show route

# Capture network traffic (built-in packet capture)
netsh trace start capture=yes tracefile=C:\evidence\trace.etl
netsh trace stop
```

icacls and takeown (File Permissions)

```
# View permissions on a file/directory
icacls C:\Windows\System32\sethc.exe

# Take ownership of a file
takeown /f C:\path\to\file

# Grant full control to Administrators
icacls C:\path\to\file /grant Administrators:F
```

```
# Remove all permissions for a user
icacls C:\path\to\file /remove "DOMAIN\baduser"

# Reset permissions to inherited defaults
icacls C:\path\to\directory /reset /t /c
```

secedit (Security Policy)

```
# Export current security policy
secedit /export /cfg C:\baseline\secpol.cfg

# Analyze against a template
secedit /analyze /db C:\temp\analysis.sdb /cfg C:\baseline\secpol.cfg

# Apply a security template
secedit /configure /db C:\temp\apply.sdb /cfg C:\hardened\secpol.cfg
```

WMIC (Windows Management Instrumentation CLI)

WMIC is deprecated but still present on all Windows versions through Server 2022. Extremely useful for quick enumeration.

```
# ---- Process Enumeration ----
wmic process list brief
wmic process where "name='powershell.exe'" get ProcessId,CommandLine

# ---- Startup Programs ----
wmic startup list full

# ---- Installed Software ----
wmic product get Name,Version,Vendor

# ---- Services ----
wmic service where "state='running'" get Name,DisplayName,PathName,StartMode

# ---- Find services running as specific accounts ----
wmic service where "StartName!='LocalSystem'" get Name,StartName,PathName

# ---- Shares ----
wmic share get Name,Path,Description

# ---- User Accounts ----
wmic useraccount get Name,SID,Status,Disabled,PasswordRequired

# ---- Network ----
wmic nicconfig where "IPEnabled=True" get Description,IPAddress,MACAddress

# ---- OS Info ----
wmic os get Caption,Version,BuildNumber,OSArchitecture,LastBootUpTime

# ---- Logical Disks ----
wmic logicaldisk get Name,Description,FileSystem,Size,FreeSpace

# ---- Remote execution (if WMI is accessible) ----
wmic /node:"TARGET_IP" /user:"admin" process call create "cmd /c ipconfig > C:\out.txt"
```

perfmon and logman (Performance Monitoring)

Built-in performance data collection. Useful for detecting resource abuse and establishing baselines.

```
# ---- Quick Performance Checks ----
# CPU and memory per process (like Task Manager in CLI)
Get-Process | Sort-Object CPU -Descending | Select-Object -First 15 Name,Id,CPU,WS

# ---- Data Collector Sets with logman ----
# Create a collector for CPU, memory, disk, network
logman create counter "CCDC_Baseline" -c "\Processor(_Total)\% Processor Time" "\Memory\Available MBytes"
↪  "\PhysicalDisk(_Total)\% Disk Time" "\Network Interface(*)\Bytes Total/sec" -si 5 -f csv -o
↪  C:\PerfLogs\baseline.csv

# Start collection
logman start "CCDC_Baseline"

# Stop collection
logman stop "CCDC_Baseline"

# Delete collector set
logman delete "CCDC_Baseline"

# List all collector sets
logman query

# ---- typeperf (real-time counter output) ----
# Watch CPU in real time
typeperf "\Processor(_Total)\% Processor Time" -si 2

# Watch specific process memory
typeperf "\Process(svchost)\Working Set" -si 5

# List all available counters (huge list)
typeperf -q > C:\counters.txt
```

Windows Event Forwarding (WEF)

Built-in centralized log collection. Forwards events from member machines to a collector without third-party agents.

```
# ---- On the Collector (your monitoring box) ----
# Enable the Windows Event Collector service
wecutil qc /q

# Create a subscription (XML-based)
# Save this as C:\WEF\subscription.xml, then:
wecutil cs C:\WEF\subscription.xml

# List subscriptions
wecutil es

# Check subscription status
wecutil gs "YourSubscriptionName"

# Retry subscription
wecutil rs "YourSubscriptionName"

# ---- On Source Machines (via GPO or manually) ----
# Enable WinRM
winrm quickconfig -q

# Add the collector to the event log readers group
# Or configure via GPO:
#   Computer Config > Admin Templates > Windows Components >
#   Event Forwarding > Configure target Subscription Manager
#   Value: Server=http://COLLECTOR_FQDN:5985/wsman/SubscriptionManager/WEC
```

```
# ---- Verify events are flowing ----
# On collector, check Forwarded Events log
wevtutil qe ForwardedEvents /c:5 /f:text
```

Windows Sandbox

Built-in disposable VM (Windows 10/11 Pro and Enterprise, Server 2019+). Opens a clean Windows instance that is destroyed on close. Useful for testing suspicious files, running untrusted tools, or testing configurations.

```
# Enable Windows Sandbox (requires reboot)
Enable-WindowsOptionalFeature -FeatureName "Containers-DisposableClientVM" -Online

# Launch (after reboot)
# Start menu > "Windows Sandbox" or:
WindowsSandbox.exe

# Automate with a .wsb config file (save as test.wsb, double-click):
# <Configuration>
#   <MappedFolders>
#     <MappedFolder>
#       <HostFolder>C:\Tools</HostFolder>
#       <SandboxFolder>C:\Users\WDAGUtilityAccount\Desktop\Tools</SandboxFolder>
#       <ReadOnly>true</ReadOnly>
#     </MappedFolder>
#   </MappedFolders>
#   <LogonCommand>
#     <Command>explorer.exe C:\Users\WDAGUtilityAccount\Desktop\Tools</Command>
#   </LogonCommand>
# </Configuration>
```

Third-Party Security Tools

Chainsaw (Event Log Analysis)

What: Fast Windows event log analysis tool. Searches EVTX files using Sigma detection rules and built-in logic to find evidence of attacks. Written in Rust, extremely fast.

Download: https://github.com/WithSecureLabs/chainsaw/releases

```
# Hunt through event logs using Sigma rules
chainsaw.exe hunt C:\Windows\System32\winevt\Logs\ -s sigma/ --mapping mappings/sigma-event-logs-all.yml

# Search for specific string in event logs
chainsaw.exe search "mimikatz" C:\Windows\System32\winevt\Logs\

# Dump all event logs to readable format
chainsaw.exe dump C:\Windows\System32\winevt\Logs\ --json
```

Hayabusa (Windows Event Log Fast Forensics)

What: Windows event log fast forensics timeline generator. Processes EVTX files with 4000+ Sigma-compatible detection rules. Generates a timeline of suspicious events.

Download: https://github.com/Yamato-Security/hayabusa/releases

```
# Generate timeline from live system
hayabusa.exe csv-timeline -d C:\Windows\System32\winevt\Logs\ -o timeline.csv

# Generate with minimum alert level
hayabusa.exe csv-timeline -d C:\Windows\System32\winevt\Logs\ -o timeline.csv -m medium

# Update detection rules
hayabusa.exe update-rules
```

DeepBlueCLI (PowerShell Threat Hunting)

What: PowerShell module for threat hunting via Windows Event Logs. Detects attack techniques including password spraying, user creation, service installation, and encoded PowerShell.

Download: https://github.com/sans-blue-team/DeepBlueCLI

```
# Analyze Security log
.\DeepBlue.ps1 -log Security

# Analyze System log
.\DeepBlue.ps1 -log System

# Analyze saved EVTX file
.\DeepBlue.ps1 C:\evidence\security.evtx
```

PingCastle (Active Directory Security Audit)

What: Audits Active Directory security health. Generates a risk score and detailed report covering trust relationships, privileged accounts, stale objects, GPO issues, and known AD attack paths. Single EXE, no installation.

Download: https://www.pingcastle.com/download

```
# Run health check (produces HTML report)
PingCastle.exe --healthcheck

# Run specific scanner
PingCastle.exe --scanner laps_bitlocker
PingCastle.exe --scanner share
```

HardeningKitty (Windows Hardening Audit)

What: PowerShell module that audits Windows configuration against CIS, DISA STIG, and Microsoft baselines. Can also apply hardening settings automatically.

Download: https://github.com/scipag/HardeningKitty

```
# Import module
Import-Module .\HardeningKitty.psm1

# Audit mode (read-only, generates report)
Invoke-HardeningKitty -Mode Audit -Log

# Apply hardening (changes settings)
Invoke-HardeningKitty -Mode HailMary
```

Loki (IOC Scanner)

What: Simple IOC and YARA scanner for endpoints. Scans for known malware hashes, filenames, C2 indicators, and YARA rule matches. Lightweight, portable.

Download: https://github.com/Neo23x0/Loki/releases

```
# Run full scan
loki.exe

# Scan specific directory
loki.exe -p C:\Users

# Update signatures
loki.exe --update
```

Eric Zimmerman's Tools (Forensic Analysis)

What: Collection of 30+ forensic analysis tools for parsing Windows artifacts: registry hives, NTFS metadata, prefetch files, shimcache, amcache, event logs, and more.

Download: https://ericzimmerman.github.io/#!index.md

Key tools:

Tool	Purpose
`MFTECmd.exe`	Parse NTFS Master File Table (find deleted files)
`PECmd.exe`	Parse Prefetch files (program execution history)
`RECmd.exe`	Parse registry hives (find persistence, config)
`ShellBagsExplorer`	Parse ShellBags (directory browsing history)
`AmcacheParser.exe`	Parse Amcache (program installation/execution)
`EvtxECmd.exe`	Parse EVTX event logs with timeline output
`Timeline Explorer`	GUI for viewing CSV timelines from other tools

DSInternals (AD Password Auditing)

What: PowerShell module for AD security auditing. Can check AD passwords against breached password lists, test password quality, and export password hashes for offline auditing.

Download: `Install-Module DSInternals -Force`

```
# Test AD passwords against a breach list (like HaveIBeenPwned)
$passwords = Get-ADReplAccount -All -Server DC01.domain.local
Test-PasswordQuality -Account $passwords -WeakPasswordHashesFile pwned-passwords-ntlm.txt
```

BloodHound (AD Attack Path Analysis)

What: Maps Active Directory trust relationships, group memberships, and permissions to visualize attack paths from any user to Domain Admin. Defenders use it to find and eliminate these paths before attackers do.

Download: https://github.com/SpecterOps/BloodHound/releases

```
# Collect AD data (run SharpHound collector)
.\SharpHound.exe -c All

# Import the ZIP into BloodHound GUI and analyze paths
```

Microsoft Attack Surface Analyzer

What: Compares system state before and after software installation or configuration changes. Takes snapshots and diffs them to show what changed (files, ports, services, registry, users, certificates).

Download: https://github.com/microsoft/AttackSurfaceAnalyzer/releases

```
# Take baseline snapshot
asa.exe collect -a

# Make changes, then take second snapshot
asa.exe collect -a

# Generate diff report
asa.exe export-collect
```

KAPE (Kroll Artifact Parser and Extractor)

What: Forensic triage collection tool. Rapidly collects forensic artifacts (event logs, registry hives, prefetch, browser history, MFT, etc.) from live systems into a portable package for analysis. Pairs with Eric Zimmerman's tools for parsing.

Download: https://www.kroll.com/en/services/cyber-risk/incident-response-litigation-support/kroll-artifact-parser-extractor-kape

```
# Collect common forensic artifacts from C: drive
kape.exe --tsource C: --tdest C:\KAPEOutput --target !SANS_Triage

# Collect and process with built-in modules
kape.exe --tsource C: --tdest C:\KAPEOutput --target !SANS_Triage --mdest C:\KAPEOutput\Modules --module
↪  !EZParser

# Collect specific targets
kape.exe --tsource C: --tdest C:\KAPEOutput --target EventLogs,RegistryHives,Prefetch,WebBrowsers

# Collect from remote share (mount evidence drive)
kape.exe --tsource \\server\share --tdest C:\KAPEOutput --target !SANS_Triage

# Key target groups:
#   !SANS_Triage   -- comprehensive artifact collection
#   EventLogs      -- Windows event logs
#   RegistryHives  -- SAM, SECURITY, SYSTEM, SOFTWARE, ntuser.dat
#   Prefetch       -- program execution history
#   $MFT           -- NTFS master file table
#   WebBrowsers    -- browser history, cookies, cache
#   Antivirus      -- AV logs (Defender, Symantec, etc.)
```

CimSweep (Agentless Remote IR via WMI/CIM)

What: PowerShell framework for remote incident response using WMI/CIM. No agent installation required on targets – works through existing Windows management infrastructure. Ideal for sweeping multiple machines quickly.

Source: https://github.com/PowerShellMafia/CimSweep

```
# Import the module
Import-Module CimSweep

# Create CIM sessions to remote targets
$s = New-CimSession -ComputerName DC01,WEB01,FILE01

# ---- Sweep for persistence mechanisms ----
Get-CSStartMenuEntry -CimSession $s
Get-CSRegistryAutoStart -CimSession $s
```

```
Get-CSScheduledTaskEntry -CimSession $s
Get-CSWmiPersistence -CimSession $s

# ---- Sweep for services ----
Get-CSService -CimSession $s | Where-Object {$_.PathName -notlike "*Windows*"}

# ---- Sweep for processes ----
Get-CSProcess -CimSession $s | Where-Object {$_.ParentProcessId -ne 0}

# ---- Sweep for network connections ----
Get-CSTCPConnection -CimSession $s

# ---- File system sweep ----
Get-CSDirectoryListing -DirectoryPath "C:\Users\Public" -CimSession $s
Get-CSRegistryValue -Hive HKLM -SubKey "SOFTWARE\Microsoft\Windows\CurrentVersion\Run" -CimSession $s
```

Purple Knight (Active Directory Security Assessment)

What: Free AD security assessment tool from Semperis. Runs 130+ security checks against Active Directory covering delegation issues, account security, Group Policy, Kerberos, and AD infrastructure weaknesses. Produces an executive-level report with risk scores and remediation guidance.

Download: https://www.purple-knight.com/

```
# Run as Domain Admin on a domain-joined machine
# 1. Extract the ZIP
# 2. Run PurpleKnight.exe
# 3. Select your forest/domain
# 4. Choose indicator categories to scan:
#    - Account Security
#    - AD Delegation
#    - AD Infrastructure
#    - Group Policy Security
#    - Kerberos Security
# 5. Review HTML/PDF report with risk scores per finding
#
# Key findings it catches:
#   - Kerberoastable accounts with weak passwords
#   - Unconstrained delegation enabled on computers
#   - AdminSDHolder permission changes
#   - DCSync permissions granted to non-DC accounts
#   - GPO permissions allowing privilege escalation
#   - Print Spooler running on DCs
#   - Stale AD objects and dormant admin accounts
```

PlumHound (BloodHound Reporting for Blue Teams)

What: Reporting engine that sits on top of BloodHound's Neo4j database. Generates actionable blue team reports – instead of "here's how attackers can reach Domain Admin," PlumHound produces "here are the 10 accounts you should fix first." Turns attack graphs into prioritized remediation lists.

Source: https://github.com/PlumHound/PlumHound

```
# Install (requires Python 3, BloodHound/Neo4j running)
git clone https://github.com/PlumHound/PlumHound.git
cd PlumHound
pip3 install -r requirements.txt

# Run default blue team reports
python3 PlumHound.py -x tasks/default.tasks -s "bolt://localhost:7687" -u neo4j -p "password"
```

```
# Run specific task
python3 PlumHound.py --easy -s "bolt://localhost:7687" -u neo4j -p "password"

# Key reports generated:
#   - Kerberoastable users ranked by privilege
#   - Users with paths to Domain Admin
#   - Computers with unconstrained delegation
#   - Group nesting issues (circular, excessive depth)
#   - Admin accounts not in Protected Users group
#   - Stale computer accounts with elevated privileges
```

Microsoft Security Compliance Toolkit (LGPO)

What: Official Microsoft tool for applying security baselines via local Group Policy. LGPO.exe imports/exports local policy settings without requiring Active Directory GPOs. The Security Compliance Toolkit also includes recommended baseline GPOs for each Windows version.

Download: https://www.microsoft.com/en-us/download/details.aspx?id=55319

```
# Export current local policy
LGPO.exe /b C:\Backups\GPO_Backup

# Apply a Microsoft security baseline
# (extract baseline ZIP, find the GPO folders)
LGPO.exe /g "C:\Baselines\Win2022-Member-Server\{GUID}"

# Import a specific registry policy file
LGPO.exe /m C:\Baselines\registry.pol

# Parse a policy file to text (review before applying)
LGPO.exe /parse /m C:\Baselines\registry.pol > C:\review.txt

# Key baselines available:
#   Windows Server 2022 (Member Server and DC)
#   Windows Server 2019 (Member Server and DC)
#   Windows 11 / Windows 10
#   Microsoft Edge
#   Microsoft 365 Apps
```

Windows Quick Reference Commands

Incident Response One-Liners

```
# Who is logged in right now?
query user

# What is running and connecting out?
Get-NetTCPConnection -State Established |
    Select LocalPort,RemoteAddress,RemotePort,OwningProcess |
    ForEach-Object { $p = Get-Process -Id $_.OwningProcess -ErrorAction SilentlyContinue;
    $_ | Add-Member -NoteProperty ProcessName -Value $p.ProcessName -PassThru } |
    Format-Table -Auto

# Find files modified in the last hour
Get-ChildItem C:\ -Recurse -File -ErrorAction SilentlyContinue |
    Where-Object {$_.LastWriteTime -gt (Get-Date).AddHours(-1)} |
    Select FullName,LastWriteTime | Sort LastWriteTime -Desc | Select -First 50

# Find unsigned executables in common malware drop locations
Get-ChildItem C:\Users,C:\Windows\Temp,C:\ProgramData -Recurse -Include *.exe,*.dll -ErrorAction
↪  SilentlyContinue |
```

```
    Get-AuthenticodeSignature | Where-Object {$_.Status -ne "Valid"} |
    Select Path,Status

# Check for hidden/system files in user directories
Get-ChildItem C:\Users -Recurse -Force -ErrorAction SilentlyContinue |
    Where-Object {$_.Attributes -match "Hidden"} | Select FullName

# Export all autorun registry keys
Get-ItemProperty "HKLM:\Software\Microsoft\Windows\CurrentVersion\Run" 2>$null
Get-ItemProperty "HKCU:\Software\Microsoft\Windows\CurrentVersion\Run" 2>$null
Get-ItemProperty "HKLM:\Software\Microsoft\Windows\CurrentVersion\RunOnce" 2>$null

# List all installed hotfixes
Get-HotFix | Sort InstalledOn -Descending | Select HotFixID,InstalledOn,Description

# Check for WMI persistence (event subscriptions)
Get-WMIObject -Namespace root\Subscription -Class __EventFilter
Get-WMIObject -Namespace root\Subscription -Class __EventConsumer
Get-WMIObject -Namespace root\Subscription -Class __FilterToConsumerBinding

# Check for alternate data streams in a directory
Get-ChildItem -Path C:\Users\Public -Recurse | ForEach-Object {
    Get-Item $_.FullName -Stream * | Where-Object Stream -ne ':$DATA'
}
```

Active Directory Quick Commands

```
# Find accounts with "Password Never Expires"
Get-ADUser -Filter {PasswordNeverExpires -eq $true} -Properties PasswordNeverExpires |
    Select Name,PasswordNeverExpires

# Find accounts that haven't logged in for 30 days
$cutoff = (Get-Date).AddDays(-30)
Get-ADUser -Filter {LastLogonDate -lt $cutoff} -Properties LastLogonDate |
    Select Name,LastLogonDate

# Find all accounts with SPN set (Kerberoastable)
Get-ADUser -Filter {ServicePrincipalName -like "*"} -Properties ServicePrincipalName |
    Select Name,ServicePrincipalName

# Find accounts with no Kerberos pre-auth (AS-REP roastable)
Get-ADUser -Filter {DoesNotRequirePreAuth -eq $true} |
    Select Name,DoesNotRequirePreAuth

# Reset all user passwords in an OU
Get-ADUser -SearchBase "OU=Users,DC=domain,DC=local" -Filter * |
    Set-ADAccountPassword -NewPassword (ConvertTo-SecureString "N3wP@ss!" -AsPlainText -Force) -Reset

# Disable all accounts except specific ones
$keep = @("Administrator","svcadmin","scoring")
Get-ADUser -Filter {Enabled -eq $true} |
    Where-Object {$keep -notcontains $_.SamAccountName} |
    Disable-ADAccount

# Force GPO update on all domain computers
Get-ADComputer -Filter * | ForEach-Object {
    Invoke-GPUpdate -Computer $_.Name -Force -RandomDelayInMinutes 0
}

# Export all GPOs
Get-GPO -All | ForEach-Object {
```

```
    Backup-GPO -Guid $_.Id -Path C:\baseline\gpo-backups
}
```

Linux Tools Reference

Process Monitoring

htop / atop

What: Interactive process viewers. `htop` is a colorful replacement for `top`. `atop` records historical system load and per-process resource usage.

```
# Install
sudo apt install htop atop    # Debian/Ubuntu
sudo yum install htop atop    # RHEL/CentOS

# Usage
htop                          # Interactive process viewer
atop                          # System + process monitor with history
atop -r /var/log/atop/atop_20260316   # Replay historical data
```

htop key bindings: `F5` = tree view, `F6` = sort, `F4` = filter by name, `F9` = kill, `t` = tree toggle, `H` = hide user threads.

ps and pstree

What: Process listing and tree visualization. Available on every Linux system.

```
# Full process listing with command lines
ps auxf

# Show all processes for a specific user
ps -u www-data

# Process tree
pstree -p         # With PIDs
pstree -pa        # With PIDs and arguments

# Find processes by name
ps aux | grep sshd
pgrep -la sshd    # Cleaner alternative
```

strace

What: Trace system calls made by a process. Shows every file open, read, write, network operation, and signal.

```
# Install
sudo apt install strace       # Debian/Ubuntu
sudo yum install strace       # RHEL/CentOS

# Trace a running process
sudo strace -fp $(pgrep suspicious_proc) -e trace=network,file

# Trace a new command
strace -f -e trace=network curl http://example.com

# Common filters
-e trace=file          # File operations only
-e trace=network       # Network calls only
-e trace=process       # Process creation (fork/exec)
-e trace=write         # Write calls only
```

lsof

What: Lists open files, network connections, sockets, and pipes for all processes. One of the most versatile tools on Linux.

```
# All network connections
sudo lsof -i

# TCP connections only, with numeric ports
sudo lsof -i TCP -n -P

# Who is listening on port 80?
sudo lsof -i :80

# All files opened by a specific process
sudo lsof -p 1234

# All files opened by a specific user
sudo lsof -u www-data

# Find which process has a file open
sudo lsof /var/log/syslog

# Find deleted but still-open files (disk space not freed)
sudo lsof +L1
```

/proc Filesystem

What: Virtual filesystem exposing kernel and process data in real time. Available on every Linux system without installing anything.

```
# Command line of a process
cat /proc/<PID>/cmdline | tr '\0' ' '

# Actual binary (symlink -- may differ from what ps shows if replaced)
ls -la /proc/<PID>/exe

# Environment variables
cat /proc/<PID>/environ | tr '\0' '\n'

# Open file descriptors
ls -la /proc/<PID>/fd/

# Network connections for a specific process
cat /proc/<PID>/net/tcp
```

pspy (Unprivileged Process Monitor)

What: Monitors new process execution without root by scanning /proc. Detects cron jobs, scripts, and hidden processes that only run briefly.

Download: https://github.com/DominicBreuker/pspy/releases

```
./pspy64             # Monitor all new processes
./pspy64 -i 100      # Scan interval 100ms for faster detection
```

Network Monitoring

ss (Socket Statistics)

What: Modern replacement for `netstat`. Shows TCP, UDP, UNIX sockets with process info. Faster and more feature-rich.

```
# All listening TCP ports with process names
ss -tlnp

# All established connections
ss -tnp

# All listening + established
ss -tunap

# Connections to a specific port
ss -tnp dst :443

# Connections from a specific IP
ss -tnp src 10.0.0.50

# Summary statistics
ss -s
```

tcpdump

What: Command-line packet capture. Captures and displays network traffic in real time or saves to pcap files.

```
# Install
sudo apt install tcpdump      # Debian/Ubuntu
sudo yum install tcpdump      # RHEL/CentOS

# Capture all traffic on eth0
sudo tcpdump -i eth0

# Capture specific port, write to file
sudo tcpdump -i eth0 port 22 -w /tmp/ssh-traffic.pcap

# Capture traffic from specific host
sudo tcpdump -i eth0 host 10.0.0.100

# Capture and display packet contents
sudo tcpdump -i eth0 -A -s 0 port 80

# Capture DNS queries
sudo tcpdump -i eth0 port 53

# Read a pcap file
tcpdump -r /tmp/capture.pcap

# Common filters
sudo tcpdump -i eth0 'tcp[tcpflags] & (tcp-syn) != 0'   # SYN packets
sudo tcpdump -i eth0 'port 80 and host 10.0.0.1'        # Combined
```

iftop / nethogs / nload

What: Real-time bandwidth monitoring tools. `iftop` shows connections by bandwidth, `nethogs` shows bandwidth per process, `nload` shows per-interface throughput.

```
# Install
sudo apt install iftop nethogs nload

# Bandwidth by connection (like TCPView for bandwidth)
sudo iftop -i eth0

# Bandwidth by process (which process is using network)
```

```
sudo nethogs eth0

# Per-interface throughput graph
nload eth0
```

conntrack

What: View and manage the kernel's connection tracking table (netfilter/iptables state table).

```
# Install
sudo apt install conntrack     # Debian/Ubuntu

# List all tracked connections
sudo conntrack -L

# Count connections
sudo conntrack -C

# Watch new connections in real time
sudo conntrack -E

# Filter by source
sudo conntrack -L -s 10.0.0.100
```

Autostart and Persistence Auditing

Linux has no single tool like Autoruns. You must check multiple locations:

```
# ---- Systemd Services ----
systemctl list-unit-files --type=service --state=enabled
systemctl list-timers --all

# ---- Cron Jobs (all users) ----
for user in $(cut -f1 -d: /etc/passwd); do
    echo "=== $user ==="; crontab -l -u "$user" 2>/dev/null
done
cat /etc/crontab
ls -la /etc/cron.d/ /etc/cron.daily/ /etc/cron.hourly/ /etc/cron.weekly/ /etc/cron.monthly/

# ---- Init Scripts ----
ls -la /etc/init.d/
cat /etc/rc.local 2>/dev/null

# ---- Shell Profiles (backdoor commands) ----
grep -r "." /etc/profile.d/
for user_home in /home/* /root; do
    echo "=== $user_home ==="
    grep -n "." "$user_home/.bashrc" "$user_home/.bash_profile" \
        "$user_home/.profile" 2>/dev/null | grep -v "^#"
done

# ---- SSH Authorized Keys ----
find / -name authorized_keys 2>/dev/null -exec echo "=== {} ===" \; -exec cat {} \;

# ---- At Jobs ----
atq
ls -la /var/spool/at/

# ---- User Systemd Services ----
find /home -path "*/.config/systemd/user/*.service" 2>/dev/null
```

```
# ---- Kernel Modules ----
lsmod
cat /etc/modules 2>/dev/null
ls /etc/modules-load.d/
```

Security Auditing

lynis

What: Comprehensive system auditing tool. Scans the system and produces a hardening report with a score and specific recommendations.

```
# Install
sudo apt install lynis        # Debian/Ubuntu
sudo yum install lynis        # RHEL/CentOS (EPEL)

# Run full audit
sudo lynis audit system

# Run specific test group
sudo lynis audit system --tests-from-group "firewalls"

# Key output files
/var/log/lynis.log            # Full log
/var/log/lynis-report.dat     # Machine-readable report
```

chkrootkit / rkhunter

What: Rootkit detection scanners. Check for known rootkit signatures, suspicious files, and system binary modifications.

```
# Install
sudo apt install chkrootkit rkhunter

# Run chkrootkit
sudo chkrootkit

# Run rkhunter
sudo rkhunter --update        # Update signatures
sudo rkhunter --check         # Full scan
sudo rkhunter --check --sk    # Skip keypress prompts
```

AIDE (Advanced Intrusion Detection Environment)

What: File integrity monitoring. Creates a database of file hashes and permissions, then detects changes.

```
# Install
sudo apt install aide

# Initialize database (do this on a clean system)
sudo aideinit

# Check for changes
sudo aide --check

# Update database after legitimate changes
sudo aide --update
```

auditd / ausearch / aureport

What: Linux kernel audit framework. Logs syscalls, file access, user commands, and security events.

```
# Install
sudo apt install auditd

# Add audit rules
sudo auditctl -w /etc/passwd -p wa -k passwd_changes
sudo auditctl -w /etc/shadow -p wa -k shadow_changes
sudo auditctl -a always,exit -F arch=b64 -S execve -F euid=0 -k root_commands

# Search audit logs
sudo ausearch -k passwd_changes          # By key
sudo ausearch -ui 0 -sc execve           # Root command execution
sudo ausearch -ts recent                  # Last 10 minutes

# Generate reports
sudo aureport --summary
sudo aureport --auth                     # Authentication events
sudo aureport --login                    # Login events
sudo aureport --file                     # File access events
```

ClamAV

What: Open-source antivirus. Scans files for known malware signatures.

```
# Install
sudo apt install clamav clamav-daemon

# Update signatures
sudo freshclam

# Scan a directory
clamscan -r /home/
clamscan -r --infected --remove /tmp/    # Remove infected files

# Scan and log
clamscan -r /var/www/ -l /var/log/clamscan.log
```

File and Disk Analysis

find (Security-Relevant Searches)

```
# Find all SUID binaries (privilege escalation vectors)
find / -perm -4000 -type f 2>/dev/null

# Find all SGID binaries
find / -perm -2000 -type f 2>/dev/null

# Find world-writable files
find / -perm -0002 -type f 2>/dev/null

# Find world-writable directories
find / -perm -0002 -type d 2>/dev/null

# Find files modified in the last 24 hours
find / -mtime -1 -type f 2>/dev/null

# Find files modified in the last hour
```

```
find / -mmin -60 -type f 2>/dev/null

# Find files owned by no user (orphaned)
find / -nouser 2>/dev/null

# Find files with capabilities set
getcap -r / 2>/dev/null

# Find hidden files in web directories
find /var/www -name ".*" -type f 2>/dev/null

# Find PHP files in upload directories
find /var/www -name "*.php" -newer /var/www/index.html 2>/dev/null
```

debsums / rpm -Va (Package Integrity)

What: Verify installed package files against their expected checksums. Detects modified system binaries.

```
# Debian/Ubuntu
sudo apt install debsums
sudo debsums -c              # Show only changed files
sudo debsums -s              # Silent (errors only)

# RHEL/CentOS
sudo rpm -Va                 # Verify all packages
sudo rpm -Vf /usr/bin/ssh    # Verify specific file's package
```

rpm -Va output codes: `S` = size, `M` = permissions, `5` = MD5, `T` = mtime, `U` = owner, `G` = group.

inotifywait (Real-Time File Monitoring)

What: Watch files/directories for changes in real time. Part of `inotify-tools`.

```
# Install
sudo apt install inotify-tools

# Watch a directory for any changes
inotifywait -m -r /etc/

# Watch specific events
inotifywait -m -r -e modify,create,delete /var/www/html/

# Watch and log with timestamps
inotifywait -m -r -e modify,create,delete --timefmt '%F %T' \
    --format '%T %w%f %e' /etc/ >> /var/log/etc-changes.log &
```

strings / file / hexdump

What: Examine file contents and types without executing them.

```
# Identify file type
file suspicious_binary

# Extract printable strings (find URLs, IPs, paths)
strings suspicious_binary | grep -E "(http|https|ftp|/bin/|/tmp/)"

# Hex dump
hexdump -C suspicious_binary | head -50
xxd suspicious_binary | head -50
```

System Information

```
# ---- OS and Kernel ----
uname -a                      # Kernel version
cat /etc/os-release           # Distribution info
hostnamectl                   # Hostname, OS, kernel, architecture

# ---- Hardware ----
lscpu                         # CPU info
free -h                       # Memory usage
df -h                         # Disk usage
lsblk                         # Block devices
ip addr                       # Network interfaces

# ---- Users and Sessions ----
w                             # Who is logged in and what they're doing
who                           # Logged in users
last                          # Login history
lastb                         # Failed login history
id                            # Current user info

# ---- Network ----
ip route                      # Routing table
ip neigh                      # ARP table
cat /etc/resolv.conf          # DNS configuration
iptables -L -n -v             # Firewall rules

# ---- Services ----
systemctl list-units --type=service --state=running
systemctl list-units --type=service --state=failed
```

Remote Administration

SSH Tips for Competition

```
# Generate key pair (do this on your admin machine)
ssh-keygen -t ed25519 -f ~/.ssh/ccdc_key

# Copy key to target (faster than typing passwords each time)
ssh-copy-id -i ~/.ssh/ccdc_key user@target

# Run command on remote host
ssh user@target "systemctl status sshd"

# Copy files
scp local_file user@target:/remote/path
scp -r user@target:/remote/dir ./local_dir

# SSH tunnel (access remote web UI through SSH)
ssh -L 8080:localhost:80 user@target
# Then open http://localhost:8080 in your browser
```

tmux (Terminal Multiplexer)

What: Persistent terminal sessions that survive SSH disconnects. Essential for competition – if your SSH drops, your work continues.

```
# Install
sudo apt install tmux

# Start named session
tmux new -s hardening

# Detach (without killing session)
Ctrl+B, then D

# Reattach
tmux attach -t hardening

# List sessions
tmux ls

# Split panes
Ctrl+B, then %          # Vertical split
Ctrl+B, then "          # Horizontal split
Ctrl+B, then arrow      # Navigate panes
```

parallel-ssh (pssh)

What: Run SSH commands on multiple hosts simultaneously.

```
# Install
sudo apt install pssh

# Create hosts file
echo "user@10.0.0.10" > hosts.txt
echo "user@10.0.0.11" >> hosts.txt
echo "user@10.0.0.12" >> hosts.txt

# Run command on all hosts
parallel-ssh -i -h hosts.txt "uname -a"

# Copy file to all hosts
parallel-scp -h hosts.txt local_file /remote/path
```

Log Analysis

journalctl

What: Query the systemd journal. The primary log viewer on modern Linux systems.

```
# Follow all logs in real time
journalctl -f

# Show logs since boot
journalctl -b

# Filter by unit
journalctl -u sshd

# Filter by time range
journalctl --since "2026-03-16 08:00" --until "2026-03-16 12:00"

# Filter by priority (0=emerg through 7=debug)
journalctl -p err            # Errors and above

# Show authentication logs
```

```
journalctl _COMM=sudo
journalctl _COMM=sshd

# Output as JSON (for parsing)
journalctl -u sshd -o json-pretty

# Disk usage
journalctl --disk-usage
```

Quick Log Searches

```
# Failed SSH logins
grep "Failed password" /var/log/auth.log | tail -20
journalctl _COMM=sshd | grep "Failed"

# Successful SSH logins
grep "Accepted" /var/log/auth.log | tail -20

# Sudo usage
grep "COMMAND" /var/log/auth.log | tail -20

# User creation
grep "useradd" /var/log/auth.log

# Cron execution
grep "CRON" /var/log/syslog | tail -20

# Count failed logins by IP
grep "Failed password" /var/log/auth.log | \
    grep -oP 'from \K[\d.]+' | sort | uniq -c | sort -rn | head
```

fail2ban

What: Monitors log files and automatically bans IPs that show malicious signs (brute force, scanning).

```
# Install
sudo apt install fail2ban

# Start and enable
sudo systemctl enable --now fail2ban

# Check status
sudo fail2ban-client status
sudo fail2ban-client status sshd

# Manually ban/unban
sudo fail2ban-client set sshd banip 10.0.0.100
sudo fail2ban-client set sshd unbanip 10.0.0.100

# List all bans
sudo fail2ban-client banned
```

DNS and Network Utilities

```
# DNS lookup
dig example.com              # Full DNS query
dig @8.8.8.8 example.com    # Query specific server
```

```
dig -x 10.0.0.1             # Reverse lookup
host example.com             # Simple lookup
nslookup example.com         # Interactive DNS

# HTTP requests
curl -I https://example.com          # Headers only
curl -v https://example.com          # Verbose (see TLS handshake)
wget --spider https://example.com    # Check if URL exists

# Certificate inspection
openssl s_client -connect example.com:443 < /dev/null
openssl x509 -in cert.pem -text -noout

# Network scanning (if nmap is available)
nmap -sV -sC -p- target             # Full port scan with versions
nmap -sn 10.0.0.0/24                # Ping sweep (host discovery)
```

Advanced Tools

osquery

What: SQL-powered operating system instrumentation. Query processes, users, network, packages, autoruns, and more using familiar SQL syntax.

Install: https://osquery.io/downloads

```
# Interactive SQL shell
osqueryi

# Example queries
SELECT * FROM processes WHERE name='sshd';
SELECT * FROM listening_ports;
SELECT * FROM crontab;
SELECT * FROM logged_in_users;
SELECT * FROM suid_bin;
SELECT * FROM shell_history;
SELECT name, path FROM startup_items;
```

YARA

What: Pattern matching tool for malware identification. Write rules to match binary or text patterns in files.

```
# Install
sudo apt install yara

# Scan files against rules
yara rules.yar /path/to/scan
yara -r rules.yar /directory       # Recursive
```

Suricata (Network IDS/IPS)

What: Network intrusion detection system. Monitors traffic for known attack signatures in real time.

```
# Install
sudo apt install suricata

# Run on interface
sudo suricata -c /etc/suricata/suricata.yaml -i eth0
```

```
# Update rules
sudo suricata-update
```

Firejail (Application Sandboxing)

What: SUID sandbox that isolates applications using Linux namespaces and seccomp-bpf. Run untrusted binaries safely.

```
# Install
sudo apt install firejail

# Run a program with no network access
firejail --net=none ./suspicious_binary

# Run with filesystem restrictions
firejail --private ./suspicious_binary
```

Cross-Reference: Windows vs. Linux Equivalents

Task	Windows (Sysinternals)	Linux
Process viewer	Process Explorer	`htop`, `ps auxf`
Process tracing	Process Monitor	`strace -fp PID`
Network connections	TCPView	`ss -tunap`, `lsof -i`
Autostart audit	Autoruns	`systemctl`, cron, `/etc/init.d`
Open files/handles	`handle64.exe`	`lsof`
Remote execution	PsExec	`ssh`, `pssh`
Service control	PsService	`systemctl`
Process kill	PsKill	`kill`, `pkill`
Logged-on users	PsLoggedOn	`w`, `who`, `last`
System info	PsInfo	`uname -a`, `hostnamectl`
Disk usage	Du	`du -sh`, `ncdu`
File signatures	Sigcheck	`debsums`, `rpm -Va`
Event logging	Sysmon	`auditd`
Rootkit detection	(third-party)	`chkrootkit`, `rkhunter`
System audit	(third-party)	`lynis`
File integrity	(third-party)	`aide`, `tripwire`
Bandwidth by process	(third-party)	`nethogs`
Firewall	Windows Firewall / Portmaster	`iptables`, `ufw`, `firewalld`
Packet capture	Wireshark / netmon	`tcpdump`, `tshark`
File monitoring	(Process Monitor)	`inotifywait`
Antivirus scan	Windows Defender	`clamav`

Quick Install Cheatsheet

Debian / Ubuntu – Install All Tools

```
sudo apt update && sudo apt install -y \
    htop atop strace lsof tmux \
    tcpdump iftop nethogs nload conntrack \
    lynis chkrootkit rkhunter aide \
    auditd audispd-plugins \
    clamav clamav-daemon \
    inotify-tools debsums \
    fail2ban pssh nmap \
    net-tools curl wget openssl
```

RHEL / CentOS – Install All Tools

```
sudo yum install -y epel-release && sudo yum install -y \
    htop atop strace lsof tmux \
    tcpdump iftop nethogs nload conntrack-tools \
    lynis rkhunter aide \
    audit \
    clamav clamav-update clamd \
    inotify-tools \
    fail2ban pssh nmap \
    net-tools curl wget openssl
```

Windows – Deploy Sysinternals

```
# Download and extract suite
Invoke-WebRequest -Uri "https://download.sysinternals.com/files/SysinternalsSuite.zip" `
    -OutFile "$env:TEMP\SysinternalsSuite.zip"
Expand-Archive "$env:TEMP\SysinternalsSuite.zip" -DestinationPath "C:\Tools\Sysinternals"

# Or access directly from the web (requires internet)
\\live.sysinternals.com\tools\procexp64.exe
```